ECON 770 Recitation 12

Convergence, LLN and CLT

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THE UNIVERSITY of NORTH CAROLINA at CHAPEL HILL

University of North Carolina at Chapel Hill

Limit

To clearly define limit of sequence $\{a_i\}_{i=1}^{\infty}$, i.e. $\lim_{n\to\infty} a_n = a$, we use $\varepsilon - N$ (sometimes $\varepsilon - \delta$) language:

$$\forall\ \varepsilon > 0,\ \exists\ n_\varepsilon \in (0, \infty)\ s.t.\ \forall\ n \geq n_\varepsilon,\ |a_n - a| < \varepsilon$$

In other words, for any arbitrarily small $\varepsilon > 0$, we can find a large enough $n_\varepsilon > 0$ such that when $n > n_\varepsilon$, $|a_n - a|$ is smaller than the ε.

Convergence in Probability

- Notation: $X_n \xrightarrow{p} c$ or $\text{plim}_{n\to\infty} X_n = c$
- Definition: $X_n \xrightarrow{p} c$ iff

$$\lim_{n\to\infty} P(|X_n - c| \leq \delta) = 1 \quad \forall\, \delta > 0$$

or equivalently

$$\lim_{n\to\infty} P(|X_n - c| > \delta) = 0 \quad \forall\, \delta > 0$$

- Interpretation: For any $\varepsilon > 0$ and $\delta > 0$, there exists $N(\varepsilon, \delta) > 0$ s.t. for all $n > N$, $P(|X_n - c| > \delta) < \varepsilon$.

Convergence in MSE

- Notation: $X_n \xrightarrow{mse} c$
- Definition: $X_n \xrightarrow{mse} c$ iff

$$\mathbb{E}\left[(X_n - c)^2\right] \to 0$$

- Expand MSE:

$$\begin{aligned}
\mathbb{E}\left[(X_n - c)^2\right] &= \mathbb{E}\left[(X_n - \mathbb{E}[X_n] + \mathbb{E}[X_n] - c)^2\right] \\
&= \mathbb{E}\left[(X_n - \mathbb{E}[X_n])^2\right] + \mathbb{E}\left[(\mathbb{E}[X_n] - c)^2\right] \\
&+ 2\mathbb{E}\left[(X_n - \mathbb{E}[X_n])(\mathbb{E}[X_n] - c)\right] \\
&= \mathbb{E}\left[(X_n - \mathbb{E}[X_n])^2\right] + (\mathbb{E}[X_n] - c)^2
\end{aligned}$$

- L^p convergence: $X_n \xrightarrow{L^\alpha} X$ iff for $\alpha > 0$,

$$\mathbb{E}\left[|X_n - X|^\alpha\right] \to 0$$

Almost Sure Converge

- Notation: $X_n \xrightarrow{a.s.} c$ or $X_n \longrightarrow c$ w.p. 1
- Definition: $X_n \xrightarrow{a.s.} c$ iff

$$P(\lim_{n\longrightarrow\infty} X_n = c) = 1$$

or

$$\lim_{n\longrightarrow\infty} P(\sup_{m>n} |X_m - c| > \varepsilon) = 0 \quad \forall\ \varepsilon > 0$$

- Intuition*:

$$\text{Event } A \triangleq \{\omega \in \sigma(\Omega) : \lim_{n\longrightarrow\infty} X_n(\omega) = X(\omega)\} = \left\{\omega : \bigcap_{m=1}^{\infty} \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \left(|X_n(\omega) - X(\omega)| < \frac{1}{m}\right)\right\}$$

In other words, $\omega \in A$ means that, for any $m > 0$ (equivalently, arbitrary small $\varepsilon > 0$), there exists some k such that for any $n > k$, $|X_n(\omega) - X(\omega)| < \dfrac{1}{m}$

a.s. - Equivalent Definition?*

- Recall that we have another definition

$$\lim_{n\to\infty} P(\sup_{m>n} |X_m - X| > \varepsilon) = 0 \quad \forall\ \varepsilon > 0$$

which means that, for any given $\varepsilon > 0$, there exists N s.t. for any $n > N$, $P(\{\omega : \sup_{m>n} |X_m(\omega) - X(\omega)| > \varepsilon\}) = 0$.

- Since for any given n,

$$A_n \stackrel{\Delta}{=} \{\omega : \sup_{m>n} |X_m(\omega) - X(\omega)| > \varepsilon\} = \bigcup_{m>n} \{\omega : |X_m(\omega) - X(\omega)| > \varepsilon\}$$

and we know A_n is decreasing in n. By continuity of probability,

$$\begin{aligned} 0 = \lim_{n\to\infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) &= P\left(\bigcap_{n=1}^{\infty} \bigcup_{m>n} \{\omega : |X_m(\omega) - X(\omega)| > \varepsilon\}\right) \\ &= P\left(\lim_{n\to\infty} X_n(\omega) \neq X(\omega)\right) \end{aligned}$$

by De Morgan's Law and $P(A^c) = 1 - P(A)$

Convergence in Distribution

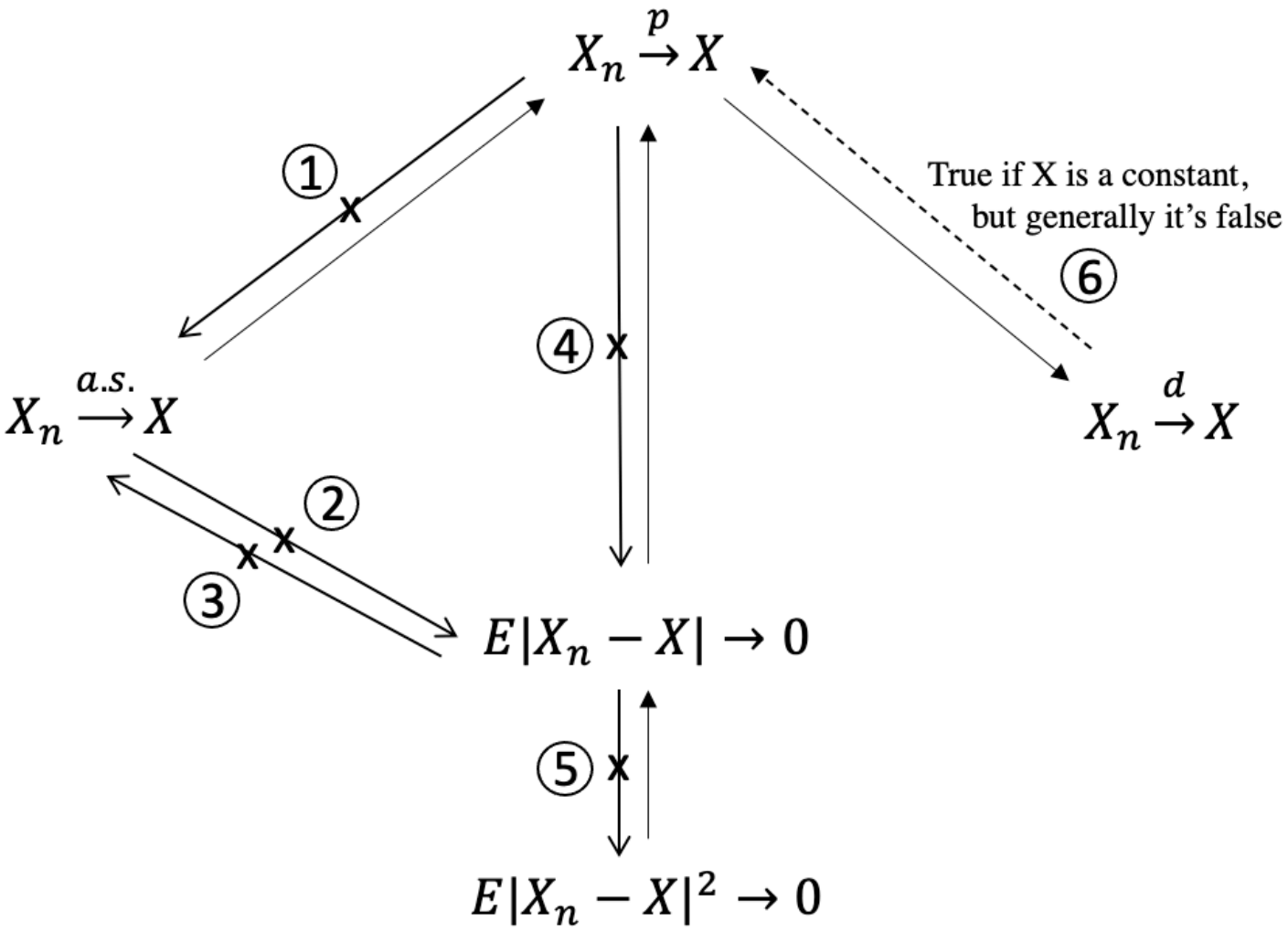
- Notation: $X_n \xrightarrow{d} X$
- Definition: $X_n \xrightarrow{d} X \sim F(\cdot)$ iff

$$P(X_n \leq x) \to P(X \leq x) \text{ at all } x \text{ at which } P(X \leq u) \text{ is continuous}$$

or

$$\lim_{n \to \infty} F_n(x) = F(x) \text{ for all } x \in \mathbb{R} \text{ at which } F \text{ is continuous}$$

Relationship Among Convergences



Counter examples are shown in the next pages

Counter Examples*

1. Let $X_1(\omega) = 1$, $X_2(\omega) = \mathbb{1}_{[0,1/2]}(\omega)$, $X_3(\omega) = \mathbb{1}_{[1/2,1]}(\omega)$, $X_4(\omega) = \mathbb{1}_{[0,1/4]}(\omega)$, $X_5(\omega) = \mathbb{1}_{[1/4,1/2]}(\omega)$, ..., then $X_n \xrightarrow{p} 0$ but it's not a.s. convergence
2. Let $P(X_n = n^2) = \frac{1}{n^2}$, $P(X_n = 0) = 1 - \frac{1}{n^2}$ and use Borel-Cantelli Lemma $X_n \xrightarrow{a.s.} 0$, but $\mathbb{E}[|X_n - 0|] = 1$ for any given n
3. Let $P(X_n = \sqrt{n}) = \frac{1}{n}$ and $P(X_n = 0) = 1 - \frac{1}{n}$
4. Let $P(X_n = n) = \frac{1}{n}$ and $P(X_n = 0) = 1 - \frac{1}{n}$
5. Let $P(X_n = \sqrt{n}) = \frac{1}{n}$ and $P(X_n = 0) = 1 - \frac{1}{n}$
6. Let $X_n \sim N(0,1)$ for all n and independent, then $X_n \xrightarrow{d} X_1$, but $X_n - X_1 \sim N(0,2)$, so let $\varepsilon = 1$ and $\lim_{n\to\infty} P(|X_n - X_1| > 1) = 2P(X_n - X_1 < -1) > 0$ for all $n \neq 1$

Weak Law of Large Numbers

Theorem

If $\{X_i\}$ is i.i.d. with mean μ and $\mathbb{E}[|X_i|] < \infty$ (or $\mathbb{E}\left[X_i^2\right] < \infty$), then

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu$$

- Proof by Chebyshev's inequality
- We say $\hat{\theta}_n$ is a <u>consistency</u> estimator of θ if $\hat{\theta}_n \xrightarrow{p} \theta$
- If $\hat{\theta}_n \xrightarrow{p} a \neq \theta$, then $\hat{\theta}_n$ is not consistent
- Example: Endogenous linear regression

$$Y = \beta_0 + \beta_1 X + \varepsilon \quad \text{where } \mathbb{E}[X\varepsilon] \neq 0$$

Strong Law of Large Numbers

Theorem

If $\{X_i\}$ is i.i.d. with mean μ and $\mathbb{E}[|X_i|] < \infty$, then

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mu$$

- Proof by Kolmogorov's in 1930 (we may assume $\mathbb{E}\left[(X_1 - \mu)^2\right] < \infty$ and $\mu = 0$ when we prove this, but it's still not straightforward)
- SLLN $\Rightarrow$ WLLN

CMT and Slutsky's Theorem

Continuous Mapping Theorem

Suppose that $g(\cdot)$ is a continuous mapping (the set of discontinuity points has zero measure), then

$$X_n \xrightarrow{d} X \Rightarrow g(X_n) \xrightarrow{d} g(X)$$

$$X_n \xrightarrow{p} X \Rightarrow g(X_n) \xrightarrow{p} g(X)$$

$$X_n \xrightarrow{\text{a.s.}} X \Rightarrow g(X_n) \xrightarrow{\text{a.s.}} g(X)$$

Slutsky (Cramér) Theorem

Suppose that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, where c is a constant, then

$$X_n + Y_n \xrightarrow{d} X + c$$

$$X_n Y_n \xrightarrow{d} Xc$$

$$X_n / Y_n \xrightarrow{d} X/c, \text{ provided that } c \neq 0$$

Some Examples

Suppose that $\{X_i, Y_i\}$ are i.i.d. and $\mathbb{E}\left[X_i^2\right], \mathbb{E}\left[Y_i^2\right] < \infty$, we can show that

- $\hat{\beta}_n = \dfrac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2} \xrightarrow{p} \dfrac{\mathbb{E}\left[X_i Y_i\right]}{\mathbb{E}\left[X_i^2\right]}$
- $\dfrac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \xrightarrow{p} Cov(X, Y)$

Lindeberg-Levy CLT

Theorem

If $X_i \sim \ i.i.d.\ (\mu, \sigma^2)$, then

$$\sqrt{n}(\bar{X} - \mu) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu) \xrightarrow{d} N(0, \sigma^2)$$

or equivalently,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right) \xrightarrow{d} N(0, 1)$$

- k-dimensional CLT: $\sqrt{n}(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} N_k(0, \boldsymbol{\Sigma})$ given i.i.d. $X_i \sim (\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- CLT family: Lyapunov CLT (independent but not identical), Lindeberg-Feller CLT (even weaker), $\alpha-$mixing CLT (Stationary and weak dependence)
- Proof by characteristic function (CF), Taylor expansion and equivalence between CF and distribution

Characteristic Function

- Definition:

$$\varphi(\lambda) = \mathbb{E}\left[e^{i\lambda X}\right]$$

  where $i = \sqrt{-1}$, which always exists
- Recall that MGF $m(\lambda) = \mathbb{E}\left[e^{\lambda X}\right]$ wherever the expectation exists
- Bounded: $|\varphi(\lambda)| \leq 1$
- Bijection: For any two random variables X_1, X_2,

$$X_1 \overset{d}{\sim} X_2 \quad \text{iff} \quad \varphi_{X_1}(\cdot) = \varphi_{X_2}(\cdot)$$

- Moment and CF: $\varphi_X^{(k)}(0) = i^k \mathbb{E}\left[X^k\right]$
- Independence (*Probability Essentials* by Jacod and Protter, Ch.14):

Corollary 14.1. *Let $X = (X_1, \ldots, X_n)$ be an $\mathbf{R}^n$-valued random variable. Then the real-valued r.v.'s $(X_j)_{1 \leq j \leq n}$ are independent if and only if*

$$\varphi_X(u_1, \ldots, u_n) = \prod_{j=1}^{n} \varphi_{X_j}(u_j) \tag{14.1}$$

Big O and Small o

Definition*

Suppose that $f(x)$ and $g(x)$ are functions defined on some unbounded subset of the positive real numbers, and $g(x) > 0$ for all large enough values of x, then

$$f(x) = O(g(x)) \quad \text{if} \quad \exists\, M > 0,\ x_0 \text{ s.t. } |f(x)| \leq Mg(x)$$

for all $x \geq x_0$

- Interpretation: $f(x)$ and $g(x)$ have the same growth rate. For example,

$$f(x) = 6x^4 - 2x^3 + 5 = O(x^4)$$

- For sequence $\{f_n\}$ and $\{g_n\}$,

$$f_n = O(g_n) \Rightarrow \frac{f_n}{g_n} = O(1) \Leftrightarrow \lim_{n\to\infty} \left|\frac{f_n}{g_n}\right| = M < \infty$$

$$f_n = o(g_n) \Rightarrow \frac{f_n}{g_n} = o(1) \Leftrightarrow \lim_{n\to\infty} \left|\frac{f_n}{g_n}\right| = 0$$

Big Op and Small op

Definition

For random variable sequence $\{X_n\}_{n\geq 1}$,

- $X_n = O_p(1)$ is defined as
$$P(|X_n| \leq M) \to 1 \quad (n \to \infty)$$
for **some** $M \in (0, \infty)$, i.e. stochastically bounded. In other words, $\forall\ \varepsilon > 0$, $\exists\ M(\varepsilon) > 0$ and $N(\varepsilon) > 0$ s.t. when $n > N$, $P(|X_n| > M(\varepsilon)) < \varepsilon$
- $X_n = o_p(1)$ is defined as
$$P(|X_n| \leq M) \to 1 \quad (n \to \infty)$$
for **any** $M \in (0, \infty)$, i.e. converge in probability. In other words, $\forall\ \varepsilon,\ \delta > 0$, $\exists\ N(\varepsilon, \delta) > 0$ s.t. when $n > N$, $P(|X_n| > \delta) < \varepsilon$

- $X_n = o_p(1) \Leftrightarrow X_n \xrightarrow{p} 0$
- $o_p(1) + o_p(1) = o_p(1)$, $o_p(1) + O_p(1) = O_p(1)$, $o_p(1)O_p(1) = o_p(1)$
- Example: $\sqrt{n}\left(\frac{\bar{X}-\mu}{\sigma}\right) \xrightarrow{d} N(0,1)$ implies $\bar{X} - \mu \to \frac{1}{\sqrt{n}}N(0, \sigma^2)$ which is stochastically bounded, therefore, $\bar{X} - \mu = O_p(1/\sqrt{n})$